Note on Quantization of Techmuller Space and CFT

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November 20, 2025

Abstract

This is my reading note for the topic of Quantization of Teichmuller Space and Conformal Field Theory. I'll maily focus on the paper by Verlinder, while adding some of the background material as well as my own understanding and comments.

Contents

1 Basic

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1.1 Quantizing the Teichmüller Space

In this subsection we follow the general philosophy of Verlinde: we aim to quantize the Teichmüller space and then identify the resulting quantum theory with the degrees of freedom of a two–dimensional CFT.

1.1.1 Definition of the Teichmüller Space

One useful way to define the Teichmüller space is as the space of all metrics on a Riemann surface Σ, modulo diffeomorphisms and Weyl transformations:

$$\mathcal{T}(\Sigma) = \frac{\text{Metrics on } \Sigma}{\text{Weyl}(\Sigma) \times \text{Diff}_0(\Sigma)}. \tag{1}$$

That is, we consider all possible metric tensors but identify those related by local Weyl rescalings and diffeomorphisms connected to the identity.

A convenient way to parametrize the metric is through zweibeine. Working in light–cone coordinates,

$$ds^2 = e^+ \otimes e^-, \tag{2}$$

where a standard choice of zweibeine is $e^+ = e^0 + e^1$ and $e^- = e^0 - e^1$.

Since 2D Lorentz transformations form a $U(1)$ group, we introduce a $U(1)$ gauge field ω, which plays the role of the spin connection. Using Cartan's structure equations, we require

$$\begin{aligned} \mathcal{D}^+ e^- &\equiv de^- + \omega \wedge e^- = 0, \\ \mathcal{D}^- e^+ &\equiv de^+ - \omega \wedge e^+ = 0. \end{aligned} \tag{3}$$

The curvature two–form is then

$$R \equiv d\omega = \Lambda\, e^{+} \wedge e^{-}. \tag{4}$$

Thus fixing the curvature R corresponds to fixing the representative within a Weyl orbit.

1.1.2 Teichmüller Space from Zweibeine

We may therefore define the Teichmüller space in terms of zweibeine as

$$\mathscr{T} = \mathscr{E}_{\mathrm{cc}}/\mathscr{G}, \qquad \mathscr{G} = \mathrm{Diff}_0 \times \mathrm{LL}, \tag{5}$$

where

- $\mathscr{E}_{\mathrm{cc}}$ is the space of zweibeine with constant curvature,
- Diff_0 is the identity component of the diffeomorphism group,
- LL denotes local Lorentz transformations, which remove the redundant freedom in choosing the zweibein.

1.1.3 Symplectic Form

Having defined the classical Teichmüller space, the next step toward quantization is to construct a natural symplectic form on $\mathscr{T}$. This will provide the phase–space structure needed for geometric quantization and eventually allow us to identify the quantized Teichmüller theory with structures appearing in 2D CFT.

2 Scratch book

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this is a scratch book file for everything